

Comparative Analysis of Deep Learning Architectures for Multi-Label Classification of *Wolffia Globosa* Conditions in Mobile Smart Farming

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Abstract—*Wolffia globosa* (duckweed) is emerging as a high-value sustainable protein source, yet its commercial viability is contingent on meeting strict food safety standards. While the crop is robust, ensuring high-quality, contaminant-free biomass is difficult for small-scale farmers who lack the expertise to visually diagnose subtle pond health issues such as nutrient imbalances or bacterial contamination. To address this, we propose a mobile-based automated diagnosis system. This study evaluates the feasibility of using deep learning models for multi-label classification of pond conditions for *Wolffia globosa* in a resource-constrained mobile environment. We curated a new dataset of pond images representing six distinct conditions, collected in a controlled experimental setup, and augmented to address class imbalances. We benchmarked seven Convolutional Neural Network (CNN) architectures, including EfficientNet (B0, B2, V2B0), ResNet50, MobileNetV3Large, VGG16, and AlexNet, using K-Fold cross-validation. Results indicate that EfficientNetB2 achieved the highest performance, with a K-Fold average accuracy of 0.7850 and a macro F1-score of 0.95. While standardized training stabilized deeper architectures like ResNet50, EfficientNetB2 remained the optimal choice, significantly outperforming legacy, non-residual architectures such as VGG16 and AlexNet. These results indicate that modern lightweight CNNs are highly effective for rapid visual diagnosis of farmer-captured images, forming the foundational core of a smart farming application.

Index Terms—Smart Farming, *Wolffia globosa*, Deep Learning, Multi-label Classification, EfficientNet, Mobile Computing.

I. INTRODUCTION

Duckweed, specifically the species *Wolffia globosa*, is gaining substantial attention in Thailand as a “superfood” due to its rapid growth cycle and high protein content [1]. However, successful cultivation for human consumption requires precise environmental management to ensure food safety. Water quality parameters, such as pH and nutrient concentration, directly impact crop health [2]. For inexperienced farmers, identifying visual signs of distress, such as “unhealthy bubbles” indicating bacterial activity or “whitening” due to fertilizer overuse, is challenging and error prone.

Manual monitoring is labor-intensive and inconsistent. While large-scale automated solutions exist, they often rely on expensive hardware or cloud-heavy computing, unsuitable for small-scale operations [3]. There is a critical need for an accessible, mobile-first solution that can diagnose pond health directly on a farmer’s consumer-grade device.

This paper presents the machine learning subsystem of our “DuWeeNext” project. We focus on the development and comparative analysis of computer vision models capable of robust multi-label classification for six specific pond conditions. This visual diagnostic tool is designed to complement a broader IoT-based water-monitoring system, providing a multi-modal approach to farm management [4]. To our knowledge, this is the first study to benchmark deep learning architectures specifically for mobile-based health monitoring of *Wolffia globosa*. Recognizing the constraints of mobile deployment, we compare legacy architectures (AlexNet, VGG16) and deeper networks (ResNet50) against more modern, efficient architectures (MobileNet, EfficientNet) to determine the optimal engine for on-device diagnosis.

II. LITERATURE REVIEW

A. Deep Learning in Plant Disease Detection

The application of Convolutional Neural Networks (CNNs) for plant disease diagnosis has become a standard approach in precision agriculture. Mohanty et al. [5] demonstrated the efficacy of deep learning by training models on the PlantVillage dataset, achieving 99.35% accuracy across 14 crop species. Their work established that CNNs could learn visual disease symptoms, such as lesions and discoloration, effectively. However, most early studies relied on heavy architectures running on server-grade hardware, presenting a barrier to adoption for smallholder farmers in remote areas.

B. Challenges in Mobile Deployment and Lightweight Models

Recent research has shifted towards enabling AI on edge devices. Howard et al. [6] introduced MobileNetV3, explicitly designed for mobile phone CPUs, by using depth-wise separable convolutions to reduce computational cost. Similarly, Tan and Le [7] proposed the EfficientNet family, using a compound scaling method to optimize accuracy while minimizing parameter count. In the context of agriculture, diverse studies have shown that these lightweight models can achieve performance comparable to heavier legacy models like VGG16 [8] but with significantly lower latency. This trend supports our hypothesis that a mobile-optimized architecture is suitable for the *Wolffia* monitoring task.

C. Visual Monitoring of Duckweed (*Wolffia globosa*)

While general plant disease detection is well-researched, specific studies on *Wolffia globosa* remain limited. Ruekaewma et al. [9] highlighted the sensitivity of *Wolffia* to nutrient variations, noting that visual symptoms often precede measurable biomass loss. Kurnia et al. [10] recently applied deep learning (StarDist) for automated duckweed counting, proving that computer vision can effectively segment minute duckweed fronds. However, their work focused on counting rather than multi-label disease classification. Our study bridges this gap by applying classification models to identify specific qualitative health conditions (e.g., Algae, Unhealthy Bubbles) rather than just quantitative growth.

To clearly situate our work within this existing landscape, Table I summarizes key studies in plant disease detection, mobile AI, and duckweed monitoring. As illustrated, a distinct gap exists for a system that combines mobile-optimized deep learning with specific qualitative disease diagnosis for *Wolffia globosa*.

III. METHODOLOGY

A. Data Collection and Experimental Setup

The primary challenge in developing a diagnostic model for *Wolffia globosa* is the scarcity of publicly available, annotated datasets. To overcome this, we constructed a custom dataset in a controlled experimental environment using circular plastic basins to simulate pond conditions. This setup allowed us to induce and isolate specific stress factors, such as excessive NPK fertilizer application, to capture distinct visual phenotypes.

Images were captured at a varying distance of approximately 3 to 12 inches (7.6 to 30.5 cm) from the water surface. This range was selected to simulate natural handheld smartphone usage while ensuring sufficient resolution to capture the granular texture details of the duckweed fronds and bubble formations.

The raw data collection yielded a total of 1,835 original images. The dataset focuses on six distinct biological and environmental classes derived from common cultivation issues, as illustrated in Fig. 2. A description of each class is detailed in Table II.

TABLE I
SUMMARY OF RELATED WORKS AND IDENTIFIED RESEARCH GAP

Author (Ref)	Domain	Methodology	Limitation/Gap
Mohanty et al. [5]	General Plant Disease Detection	Trained deep CNNs (AlexNet, GoogleNet) on the PlantVillage dataset (14 crops).	Relied on heavy, server-grade architectures; did not address resource-constrained mobile deployment.
Howard et al. [6]; Tan & Le [7]	Mobile AI Architectures	Proposed MobileNetV3 and EfficientNet with depth-wise convolutions and compound scaling.	Presented general-purpose architectures; effectiveness on microscopic aquatic plants (<i>Wolffia</i>) was not established.
Ruekaewma et al. [9]	<i>Wolffia globosa</i> Biology	Investigated the biological impact of nutrient variations on <i>Wolffia</i> growth.	Relied entirely on manual visual observation and biomass weighing, lacking automation.
Kurnia et al. [10]	<i>Wolffia</i> Automation	Used Deep Learning (StarDist) for counting duckweed fronds.	Focused strictly on quantitative counting/biomass, not qualitative health or disease classification.
Current Study	Mobile <i>Wolffia</i> Diagnosis	Benchmarked lightweight CNNs (EfficientNet) for multi-label disease classification.	Addresses the gap by providing automated, qualitative health diagnosis on mobile devices.

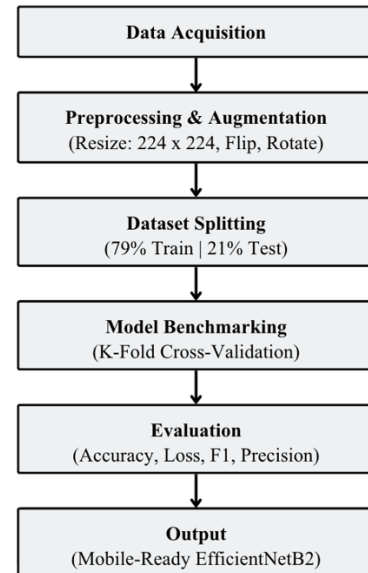


Fig. 1. The proposed methodological workflow comprising five key stages: (1) Data Acquisition from controlled basins, (2) Preprocessing & Augmentation to address class imbalance, (3) Dataset Splitting into training and test sets, (4) Model Benchmarking using K-Fold Cross-Validation, and (5) Final Evaluation to select the optimal mobile-ready architecture.

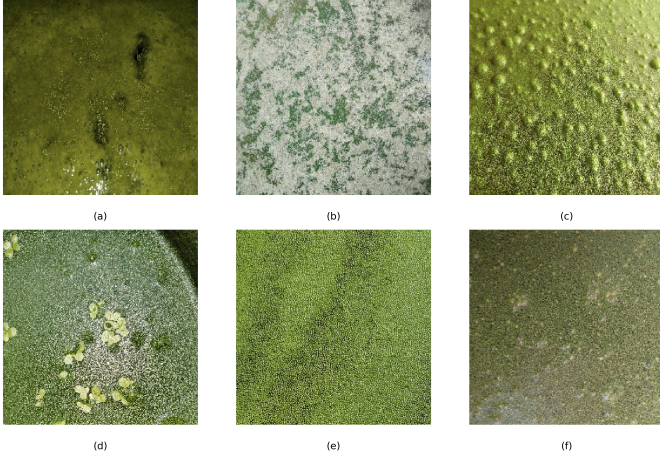


Fig. 2. Sample instances from the custom *Wolffia globosa* dataset illustrating the multi-label nature of pond health conditions. The examples exhibit both isolated and simultaneously co-occurring labels: (a) Algae and Unhealthy Bubbles, (b) Excessive NPK, (c) Healthy Bubbles, (d) Invasive Plants and Healthy Bubbles, (e) No Bubbles, and (f) Unhealthy Bubbles.

TABLE II
DATASET CLASS DESCRIPTIONS

Class Name	Description
Algae	Competitive biological contaminant
Excessive NPK	Nutrient burn/whitening
Healthy Bubbles	Normal respiration indicators
Invasive Plants	Non-duckweed flora
No Bubbles	Baseline condition
Unhealthy Bubbles	Bacterial/Water quality issues

B. Data Preprocessing and Augmentation

A significant limitation in agricultural datasets is class imbalance [10]. In our initial raw collection, the occurrence of the “unhealthy bubbles” label was rare (appearing in only 35 instances) compared to baseline conditions (which appeared in approximately 360 instances each). Training on such a skewed label distribution would lead to models biased toward the majority classes.

TABLE III
LABEL OCCURRENCES POST-AUGMENTATION (TRAIN/TEST SPLIT)

Class Name	Training Set	Testing Set	Total Occurrences
Algae	698	160	858
Excessive NPK	476	139	615
Healthy Bubbles	622	159	781
Invasive Plants	552	128	680
No Bubbles	1,318	380	1,698
Unhealthy Bubbles	352	66	418

To ensure robustness, we applied a rigorous data augmentation pipeline using the Roboflow framework. This approach synthetically expanded images containing underrepresented labels. While perfect parity is inherently difficult to achieve in multi-label datasets due to label co-occurrence, this augmen-

tation significantly increased the representation of minority labels, mitigating severe class starvation as detailed in Table III. The specific preprocessing and augmentation parameters used are detailed in Table IV.

TABLE IV
DATA AUGMENTATION AND PREPROCESSING CONFIGURATION USED TO BALANCE THE DATASET

Category	Technique	Settings/Parameter
Preprocessing	Resize Auto-Orient	Stretch to 224 x 224 pixels Applied
Augmentation	Flip Rotation Shear Brightness Exposure Noise Blur	Horizontal, Vertical 90° (Clockwise, Counter Clockwise, Upside Down) $\pm 10^\circ$ Horizontal, $\pm 10^\circ$ Vertical Between -20% and $+20\%$ Between -15% and $+15\%$ Up to 0.5% of pixels, Gaussian Blur up to 1px

Following augmentation to resolve initial class imbalances, the final dataset comprised 2,909 images. This was partitioned into a Training Set (79%, 2,302 images) and a Test Set (21%, 607 images). A static validation set was not reserved; instead, K-Fold cross-validation was utilized during the training phase.

A defining characteristic of real-world *Wolffia globosa* cultivation is that pond health conditions are rarely mutually exclusive. To accurately reflect this, our classification task was framed strictly as a multi-label problem. As detailed in Table II, the dataset spans six distinct classes. Because conditions frequently co-occur (e.g., an image exhibiting both “Algae” and “Excessive NPK”), the total number of label occurrences significantly exceeds the number of physical images.

TABLE V
MULTI-LABEL DATASET CHARACTERISTICS AND DISTRIBUTION

Metric	Training Set	Testing Set
Total Images	2,302	607
Images with 1 Label	864	227
Images with 2 Labels	1,160	335
Images with 3 Labels	278	45
Label Cardinality	1.7454	1.7002
Label Density	0.2909	0.2834

To quantify this multi-label complexity, we calculated the Label Cardinality (the average number of labels per image) and Label Density (Table V). The training set exhibits a label cardinality of 1.7454, with the majority of images containing two or more concurrent labels, necessitating architectures capable of robust multi-label inference.

C. Model Selection and Training

We selected seven Convolutional Neural Network (CNN) architectures for benchmarking:

- 1) **Legacy Models (AlexNet, VGG16, ResNet50):** These architectures were tested to establish a baseline and evaluate the “depth vs. accuracy” trade-off [5], [8].

2) **Lightweight Models (MobileNetV3Large, EfficientNetB0, B2, V2B0):** These architectures were selected specifically for their low parameter counts and suitability for resource-constrained mobile deployment [6], [7].

To ensure a fair and rigorous comparison, all models were trained using a strictly standardized hyperparameter configuration. The networks were initialized with pre-trained ImageNet weights (with the exception of the custom-built AlexNet) and fine-tuned on our dataset. The training pipeline utilized the Adam optimizer with a learning rate of 0.0005, a batch size of 32, and the binary cross-entropy loss function, which is optimal for multi-label classification tasks.

Each model was trained for exactly 30 epochs, with all input images uniformly preprocessed and resized to 224×224 pixels using the standard ImageNet utility functions. Finally, model performance was evaluated using an unstratified 5-fold cross-validation (K=5) approach. This rigorous methodology ensures the reliability of the comparative results and accurately validates the models' ability to generalize to unseen data.

IV. RESULTS

A. Comparative Accuracy and Loss

The performance of the seven models was evaluated based on K-fold Average Accuracy and K-fold Average Loss. As illustrated in Fig. 3, the EfficientNet architectures, alongside ResNet50, demonstrated the strongest performance.

- 1) EfficientNetB2 achieved the highest K-fold accuracy at 0.7850, closely followed by EfficientNetB0 (0.7754).
- 2) ResNet50 achieved a highly competitive accuracy of 0.7372, surpassing EfficientNetV2B0 (0.7341) and MobileNetV3Large (0.7180).
- 3) Legacy models without residual connections struggled: AlexNet achieved 0.5782, and VGG16 achieved the lowest accuracy at 0.4974.

B. Precision, Recall, and F1-Score

To analyze the models' ability to handle the problem's multi-label nature, we examined Precision, Recall, and F1-Score. Across the standardized evaluations, EfficientNetB2 emerged as the optimal architecture for deployment:

- 1) It achieved an exceptional Macro Average F1-Score of 0.95, matching EfficientNetB0 and slightly outperforming ResNet50 (0.94).
- 2) It demonstrated an excellent balance of Precision (0.95) and Recall (0.96). This high recall ensures that critical, less frequent issues (such as unhealthy bubbles) are rarely missed by the diagnostic system, which is paramount for early disease intervention.

V. DISCUSSION

A. Theoretical Implications

The results of this study contribute to the growing body of literature on "Green AI" and edge computing in agriculture. Theoretically, our findings challenge the assumption that

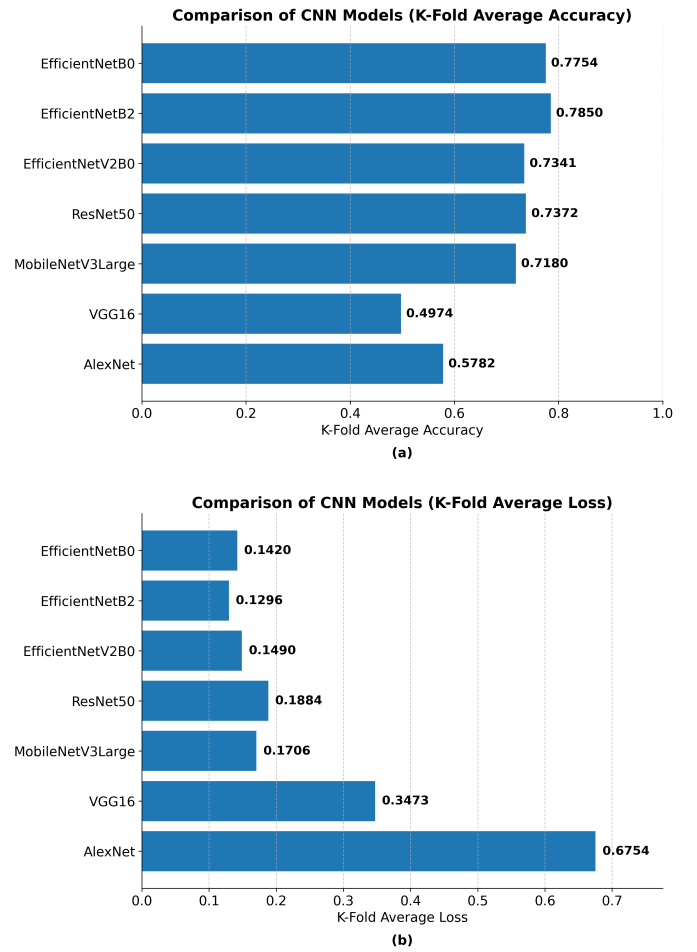


Fig. 3. Comparison of K-fold Average Accuracy (a) and Average Loss (b) across seven deep learning architectures.

deeper, parameter-heavy legacy models are strictly necessary for complex texture recognition in plants.

Architectural Efficiency vs. Raw Power: While VGG16 [8] has historically been a benchmark for image classification, its poor performance here (F1-score 0.23) highlights the phenomenon of overfitting when applying massive, non-residual architectures to specialized, limited-size datasets. Conversely, the success of EfficientNetB2 supports the theory of Compound Scaling [7]. By uniformly scaling depth, width, and resolution, EfficientNet captured the granular visual phenotypes of *Wolffia globosa* more effectively than legacy models, validating that inductive bias and architectural efficiency are more critical predictors of performance than raw parameter count alone.

Emerging Trends in Edge AI: While lightweight CNNs like EfficientNet have proven highly effective for mobile deployment, recent literature highlights several emerging architectures. Vision Transformers (ViTs) [11] have set new benchmarks in computer vision; however, they typically require substantially larger datasets for training and often present higher computational overhead for mobile edge devices compared to optimized CNNs. Similarly, YOLO (You Only Look Once)

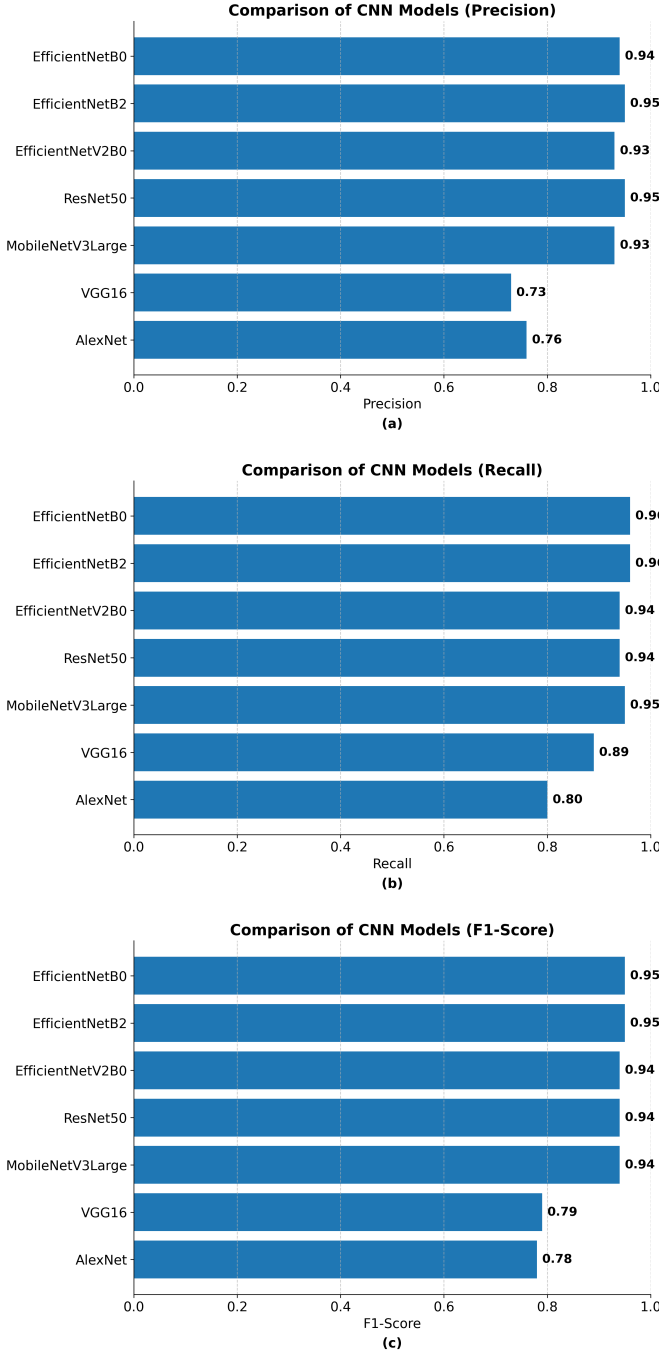


Fig. 4. Detailed performance metrics across all seven architectures. The panels illustrate: (a) Macro Average Precision showing the accuracy of positive predictions, (b) Macro Average Recall indicating the model’s ability to retrieve relevant cases, and (c) Macro Average F1-Score highlighting the overall balance.

architectures [12] are currently state-of-the-art for point detection and object bounding. While YOLO would be highly relevant for quantitative tasks such as individual duckweed frond counting, our system prioritizes holistic, image-level classification of overall water conditions. Therefore, lightweight CNNs remain a highly efficient and practical choice for this specific diagnostic scope.

Edge vs. Cloud Computing in Agriculture: Furthermore, this study offers theoretical evidence favoring Edge AI over Cloud-based approaches for rural smart farming.

- 1) **Latency:** Cloud-based systems introduce transmission latency, which is detrimental in rural areas with unstable connectivity. Our results show that lightweight architectures (EfficientNetB2) achieve high accuracy (0.95 F1-score) with low parameter counts, demonstrating the feasibility of near-zero-latency inference directly on edge devices for future mobile deployments.
- 2) **Cost & Scalability:** Cloud processing incurs recurring server costs per API call. By validating that a mobile-native architecture provides sufficient diagnostic precision, we demonstrate that transitioning to an “Edge” deployment is theoretically superior for scaling to thousands of resource-constrained farmers, as it ultimately allows processing costs to be offloaded to the user’s device rather than a central server.

B. Practical Implications

For the agricultural sector, this study provides a blueprint for deploying diagnostic tools in resource-constrained environments.

- **For Developers**

This study validates a cost-effective architectural pattern for Agritech applications. By showing that lightweight models (EfficientNetB2) achieve diagnostic precision comparable to heavy models, we demonstrate that developers can shift inference from the cloud to the edge. EfficientNetB2 drastically reduces backend operational costs (eliminating expensive GPU server fees) and simplifies infrastructure, allowing startups to scale their user base in rural markets without incurring proportional cloud computing expenses.

- **For Farmers**

The integration of this model into the DuWeeNext system effectively places an “automated agronomist” in the farmer’s pocket. It empowers inexperienced growers to identify risks, such as bacterial contamination, early, potentially increasing the yield of food-grade *Wolffia*.

- **Data Availability**

To support further research in this domain, the custom *Wolffia globosa* dataset generated during this study is available from the corresponding author upon reasonable request.

C. Limitations of the Project

While the study achieved its primary objectives, several limitations must be acknowledged:

- 1) **Experimental Environment:** Data were collected in controlled plastic basins rather than active agricultural ponds to isolate variables. Consequently, the model may require further fine-tuning and domain adaptation to handle the complex lighting conditions, natural sunlight glare, and fluctuating water turbidity found in open-field, real-world environments.
- 2) **Ground Truth Validation and Labeling Bias:** The labels were established based on a visual taxonomy derived from literature. Because this annotation process relied heavily on human visual assessment without chemical laboratory tests (e.g., water quality mass spectrometry) for biological validation, it inherently introduces human subjectivity and labeling bias. Ambiguities in visual symptoms may have led to subtle inconsistencies in the ground truth assignments.
- 3) **Resource Constraints:** This research was conducted as an undergraduate capstone project with a highly constrained budget and timeline. These constraints limited our ability to conduct longitudinal field trials. The study should therefore be viewed as a feasibility proof-of-concept rather than a commercial-ready deployment.
- 4) **Dataset Diversity and Model Generalization:** A fundamental concern in applied machine learning is a model's ability to generalize to out-of-distribution data. Because both our training and testing datasets were captured from the same controlled experimental setup and geographic location, there is an inherent risk of overfitting to this specific domain. The model's generalization capabilities must be further evaluated against images captured using different camera sensors, varying water sources, and different *Wolffia* strains.

D. Future Directions

Building upon this foundation, future work will focus on:

- 1) **Hierarchical Severity Scoring**
Moving beyond binary labels to a weighted ranking system (e.g., Excessive NPK > Unhealthy Bubbles > Healthy) to provide farmers with actionable priority scores.
- 2) **Multi-Modal IoT Fusion and Objective Validation**
We aim to correlate these visual diagnostics with time-series data from affordable IoT sensors (pH, Temperature, TDS). Fusing visual data with localized chemical sensor data will allow us to uncover latent relationships between invisible water quality changes and the onset of visible symptoms. Crucially, this sensor integration will also help mitigate the subjective biases present in our current visual labeling process by establishing a more objective, biologically validated ground truth, serving as a scalable alternative to expensive continuous laboratory testing.
- 3) **Enhancing Generalization via Cross-Domain Data Collection**
To address generalization concerns, future iterations must prioritize expanding dataset diversity. By deploy-

ing the application to a beta-testing group of farmers across diverse geographic regions and pond architectures (e.g., earth ponds, concrete tanks), a crowdsourced data pipeline can be established. Retraining the models on this highly varied, real-world data stream will mitigate environmental overfitting and ensure universal generalization.

VI. CONCLUSION

This study addresses the critical gap in automated monitoring for *Wolffia globosa*. By benchmarking seven deep learning architectures, we demonstrated that modern, lightweight CNNs were superior to legacy models for this task. Specifically, EfficientNetB2 achieved an F1-score of 0.95, demonstrating that high-accuracy diagnosis is possible on consumer-grade mobile hardware. Despite limitations in budget and field validation, this work establishes a solid baseline for AI-driven smart farming tools, offering a scalable path toward safer and more efficient superfood cultivation in Thailand.

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